Learning to Hedge with Stocks and Options

The Control Problem, Learning Method and Application Interfaces

research@strategynet.ai

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Abstract

A dealer who sells an option must finance its eventual payment while managing the risk of market moves along the way. Trading stock offsets some of that exposure; adding a traded option gives the dealer another way to manage curvature and volatility risk. Every adjustment also incurs costs. We formulate this decision as a finite-horizon stochastic control problem, specifying the market dynamics, observations, holdings, cash account and terminal loss. We explain how squared loss and conditional value at risk lead to different learning objectives, and how a neural policy learns from complete trading paths through automatic differentiation. Volt supplies the financial calculation and learner, while Jormungandr supplies the training and inference interfaces. A worked cash account, a dated decision timeline and a tail-loss diagram connect the mathematics to an application. Numerical comparisons and a synthetic learning exercise demonstrate the calculation and its application interface. The client workflow covers scenario submission, training, held-out evaluation, holdings inference and checkpoint recovery.

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1 Hedging an option book

A dealer sells a call on a stock and receives its premium. At expiry, the buyer can claim the excess of the stock price over the agreed strike. A put pays the excess of the strike over the stock price. Selling both at the same strike creates a short straddle: the dealer owes a payment when the stock finishes far enough in either direction. The premium finances the hedge, but the eventual payment and the trading needed to manage it remain uncertain.

A stock position offsets the option book's response to a small price change. Delta measures that response, in shares per option unit. Gamma measures how delta changes as the stock moves. Vega measures the response to a change in volatility, with its quoted unit specified by the application. These exposures change during the life of the book. A trader who continually adjusts a stock hedge therefore pays transaction costs while retaining exposure to movements between trading times and to changes in option prices.

A second option adds another way to manage that risk. Let L denote the liability and C a traded call used for hedging. Given their model Greeks, a delta–gamma hedge holds

$$h^C = \frac{\Gamma_L}{\Gamma_C}, \qquad h^S = \Delta_L - h^C \Delta_C. \tag{1}$$

The call offsets curvature and the stock completes the delta adjustment. The remaining vega is $\nu_L - h^C \nu_C$; its size depends on the chosen contracts. A small hedge-option gamma can also produce a very large call position. Instrument choice, position limits and trading costs matter as much as the local sensitivity calculation.

Deep hedging learns the holdings from simulated trading experience [1]. François and colleagues develop the stock-and-option setting using joint index returns and implied-volatility information [2, 3]. The policy receives current observations and existing positions, then selects the next stock and option quantities. Training rewards decisions that reduce a specified measure of the final hedging loss after costs and financing. This is useful when a simple Greek-matching rule trades too frequently, or when several instruments offer different combinations of risk reduction and execution cost. Model hedges still provide valuable comparisons: they explain what the neural policy gains or gives up as the market and cost assumptions change.

2 The market and the decision process

Write the trading dates as $t_k = k\Delta t$, $k = 0, \dots, N$. The episode ends at $T = t_N$. The policy chooses holdings at $k = 0, \dots, N-1$; the final date settles the liability and sells the hedge. Daily trading means one action per trading day. Simulation may use finer time steps to construct prices between those dates while preserving the chosen trading frequency.

Figure 1 follows one trading episode. Each decision uses the observations and holdings available at that date; the final loss supplies the training signal for all preceding actions.

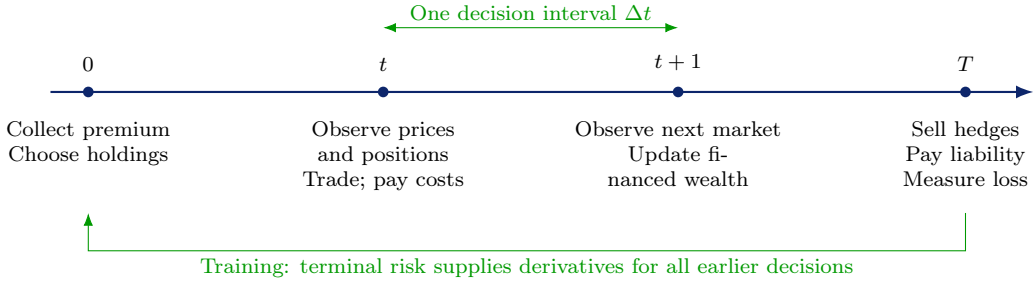


Figure 1: One episode follows the liability from sale to expiry. The policy acts at specified dates using information available at that date. Prices move between decisions; financing and trading costs enter the account before the terminal objective is calculated. Training repeats this calculation across many market paths.

A market generator describes how the external market state X_k changes:

$$X_{k+1} = F_k(X_k, \varepsilon_{k+1}), \qquad \varepsilon_{k+1} \sim P_k(\cdot \mid X_k). \tag{2}$$

The scenario probability law determines which outcomes the objective weights. A risk-neutral law supports a pricing-model benchmark; an estimated real-world law supports a hedge assessed under that return distribution. Both require consistent instrument marks and financing assumptions. Here X can contain the stock price, volatility variables, interest rates and any additional factors required by the model. The current examples treat the trader as a price taker: changing the hedge affects wealth and costs, while the external price process follows its own law. Market impact would require a transition model with the action among its inputs.

The two-day application control uses geometric Brownian motion,

$$S_{k+1} = S_k \exp\left[(r - q - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_{k+1}\right], \tag{3}$$

with independent standard-normal innovations, $\Delta t = 1/252$, $S_0 = 100$, $\sigma = 0.20$, $r = 0.03$ and $q = 0.01$. The liability pays $|S_2 - 100|$. The hedge call has strike 100 and an initial 84-trading-day maturity; its mark at each date is its Black–Scholes value using the remaining maturity and those same parameters. These choices completely specify this small market experiment.

The separate Heston study lets instantaneous variance v change randomly:

$$dS_t = \mu S_t\,dt + S_t\sqrt{v_t}\,dB_t, \tag{4}$$

$$dv_t = \kappa(\bar{v} - v_t)\,dt + \xi\sqrt{v_t}\,dW_t, \tag{5}$$

$$d\langle B, W\rangle_t = \rho\,dt. \tag{6}$$

Mean reversion has speed κ and long-run variance $\bar{v}$; ξ controls fluctuations in variance and ρ connects the two shocks. Our zero-rate Heston comparison uses $\mu = 0$, $S_0 = 100$, $v_0 = \bar{v} = 0.04$, $\kappa = 1$, $\xi = 2$, $\rho = -0.7$ and 30 daily intervals of 1/365. Its second hedge instrument is a funded variance claim. That experiment and the two-day stock-and-call control have different market and instrument definitions. Each has its own scenario and result records.

The full economic state is $(k, X_k, V_k, h_{k-1}, u_k)$: date, market variables, account wealth, existing holdings and any running contract statistic u_k. An Asian payoff, for example, requires a running average. The agent receives an observation vector $x_k = \phi_k(X_{0:k}, u_k)$ and its previous holdings. The application chooses which quantities to expose. If a latent volatility factor is omitted, the observed problem can require memory to infer it. Wealth can also be supplied as a feature when the policy should react to it. The supplied two-day policy observes scaled log spot and time remaining, plus previous holdings.

The action $h_k = (h_k^S, h_k^C) \in \mathbb{R}^2$ specifies the desired stock and option holdings. Orders are the changes $h_k - h_{k-1}$. The current continuous-output native policy permits signed quantities; adding position limits requires a specified admissible action set and its implementation. The initial holdings are zero and initial wealth is the premium. The account recurrence below, together with F_k and the terminal payoff, defines the controlled system completely.

3 Stock, option and cash positions

The account starts with wealth V_0, typically the premium collected for the liability. At each trading time t, the policy chooses stock shares h_t^S and hedge-option units h_t^C. Let $H_t = (S_t, C_t)$ be their prices, and let $h_{-1} = 0$. With proportional trading fees κ_i, the cash left after rebalancing is

$$c_t = V_t - h_t^S S_t - h_t^C C_t - \sum_{i \in \{S,C\}} \kappa_i H_t^i |h_t^i - h_{t-1}^i|. \tag{7}$$

This cash balance earns or pays interest. Under a stock dividend-reinvestment convention, the next pre-trade wealth is

$$V_{t+1} = c_t R_t + h_t^S S_{t+1} G_t + h_t^C C_{t+1}, \tag{8}$$

where $R_t = \exp(r_t \Delta t)$ and $G_t = \exp(q_t \Delta t)$. The scenario input supplies these growth factors, allowing the account convention to be checked alongside the prices. A negative cash balance represents borrowing.

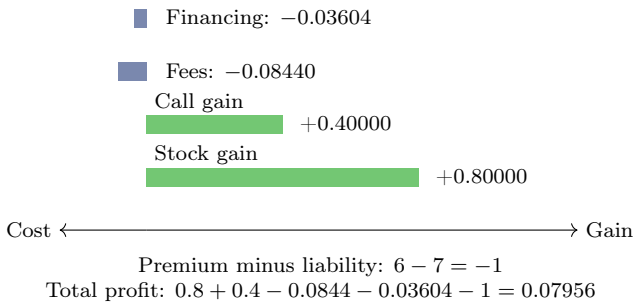
At the liability's expiry, the remaining hedge positions are sold. With closing-cost rates κ_i^{close}, the available wealth and loss are

$$V_T^{\text{liq}} = V_T - \sum_i \kappa_i^{\text{close}} H_T^i |h_{T-1}^i|, \tag{9}$$

$$E_T = L_T - V_T^{\text{liq}}. \tag{10}$$

A positive E_T is a shortfall; the seller's profit is $-E_T$. The longer-dated hedge option is sold at its expiry-date market value while the liability pays its contractual amount. Each opening, adjustment and closing fee enters the account once.

Consider a one-period example. A dealer sells one call with strike 95 for a premium of 6. The stock costs 100 and a longer-dated hedge call costs 4. Buying 0.4 shares and 0.5 hedge-option units costs 42. Stock fees of 5 basis points and option fees of 1% add 0.04, leaving cash of −36.04. Over the period, that borrowing grows by a factor of 1.001. The stock then reaches 102 and the hedge call 4.8; dividends are zero.



Premium minus liability: $6 - 7 = -1$
Total profit: $0.8 + 0.4 - 0.0844 - 0.03604 - 1 = 0.07956$

Figure 2: Sources of profit in the two-date example. Bar lengths show currency contributions from trading gains, fees and financing. The premium and liability add a net cost of one currency unit.

The profit can also be reconstructed from stock gains of 0.8, option gains of 0.4, the original premium of 6, the liability payment of 7, total fees of 0.0844 and financing of 0.03604. Both calculations give 0.07956. This independent reconstruction helps catch sign and timing errors.

Contract multipliers must be applied consistently. An exchange contract covering 100 shares corresponds to 100 option units when option prices are quoted per share. The stock quantity is already in shares. The application converts contracts to these economic quantities once, before constructing the account. The same convention carries through premiums, Greeks, fees and terminal payments. Scaling the entire example by 100 scales every cash flow and the profit by 100; mean squared loss then scales by 100^2.

4 Learning from completed trading paths

The policy is a function $h_t = \pi_\theta(x_t, h_{t-1})$, where x_t contains information available at the trading time and θ contains learned weights. Spot, time to expiry, option quotes and volatility descriptors can all be inputs. Existing holdings tell the policy how much its next decision will cost to implement. Features drawn from a full future path must be restricted to observations available at t.

For B scenario paths, one objective is mean squared hedging loss,

$$J(\theta) = \frac{1}{B} \sum_{b=1}^{B} E_T^{(b)}(\theta)^2. \tag{11}$$

This penalizes both overfunding and shortfall. Downside squared loss replaces E_T^2 by $\max(E_T, 0)^2$, emphasizing the seller's unpaid obligation. Conditional value at risk, or CVaR, concentrates on the worst losses. Its level-α representation is

$$\text{CVaR}_\alpha(E_T) = \min_\eta \left\{ \eta + \frac{\mathbb{E}[(E_T - \eta)_+]}{1 - \alpha} \right\}. \tag{12}$$

In reinforcement-learning notation, intermediate rewards can be set to zero and the final reward to $-\ell(E_T)$, with discount factor one. For squared loss, $\ell(e) = e^2$. For CVaR at a given threshold η,

$$\ell_\eta(e) = \eta + \frac{(e - \eta)_+}{1 - \alpha}. \tag{13}$$

Optimizing η alongside the policy yields the CVaR objective; the native empirical calculation evaluates the equivalent fractional-tail statistic. Funding already appears in the cash account. The trader commits at the start to the chosen terminal-risk objective and learns a policy for that objective across the full horizon. This formulation gives each action credit through its eventual effect on the account, including future trading costs.

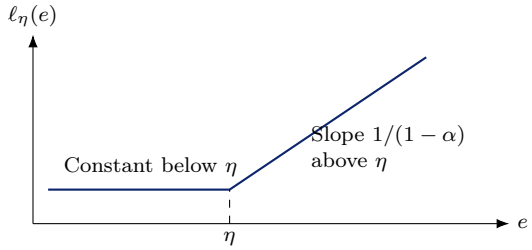


Figure 3: CVaR training puts additional weight on losses above the threshold. Increasing α concentrates the objective on a smaller, more adverse part of the loss distribution.

The native learner supports squared, downside squared, empirical CVaR and entropic objectives. Its empirical CVaR calculation weights a fractional observation at the tail cutoff when the requested tail size is nonintegral. Choice of objective changes the trading problem: a policy trained on average squared error can differ materially from one protecting against extreme shortfall.

The terminal liability and trading account determine the training signal. The learner runs each path forward, calculates the final loss, and propagates derivatives back through all holdings, fees and cash balances to the weights. An optimizer then changes those weights. Every decision date receives its learning signal from the final loss. Known model hedges provide independent performance comparisons. The supplied continuous-action application uses this direct differentiation approach within Jormungandr's learner interface.

Burn carries out automatic differentiation through the tensor calculation [7]. For exogenous simulated prices, those prices are fixed inputs during a weight update; derivatives follow the policy and the account. Absolute-value trading costs are piecewise differentiable, with the implementation's derivative convention at zero turnover. Price Greeks use differentiation with respect to financial inputs; policy training uses differentiation with respect to network weights. Volt provides both applications, with distinct inputs and outputs.

For a batch of trajectories ω_b, the actual update is

$$\widehat{J}(\theta) = \frac{1}{B} \sum_b \ell(E_T(\theta; \omega_b)), \qquad \theta \leftarrow \text{Adam}(\theta, \nabla_\theta \widehat{J}). \tag{14}$$

The implemented method is direct policy optimization through a differentiable simulator/account calculation. It uses the known derivatives of that calculation to learn continuous actions. Other RL methods can estimate a value function or a policy gradient from sampled rewards; integrating one of those engines would retain the same economic problem specification. A policy's output is the next holding, and its training target is the risk of the resulting account.

The current native policy uses a separate two-hidden-layer ReLU network at each decision date and feeds previous holdings into the next decision. It learns for the liability and scenario distribution supplied during training. At deployment, changing market observations changes the action through x_t. Supporting a broader range of books requires liability descriptors and representative portfolios in the training distribution, followed by validation across those books. Equivalent contractual decompositions should produce the same net cash flows and economic exposures under matched costs; that property can be tested directly in the account and policy inputs.

5 Training and inference through the API

A hedging application supplies its scenarios and liability, then uses the same model for training, evaluation and holdings inference. Volt evaluates the financed account and differentiates its chosen risk objective. Jormungandr manages the model and exposes the requests over HTTP [4, 5]. A registered worker retains the policy, optimizer and scenario tensors between requests. CPU, Metal and CUDA builds select where that calculation runs.

The following Python calls use `JormungandrClient`. Start the service with `VOLT_LEARNER_BIN` pointing to the compiled Volt worker. The public example `examples/native_hedging.py` launches two scenario-producing actors against it [5]. Here we follow one model through the same API, using the two-day market defined in Section 2.

Here `premium` is the initial straddle value under the stated pricing model. The default policy has two decision dates, two market features, two hedge instruments and hidden width eight. Setting the objective to `{"kind":"cvar","alpha":0.95}` selects 95% CVaR; changing the risk preference leaves the scenario submission workflow intact.

A batch supplies flattened arrays: `marks` has shape $[B, N+1, 2]$, `features` has $[B, N, 2]$ and `payoffs` has $[B]$. For each path the marks alternate stock and call prices at successive dates. Features contain scaled log spot and time remaining at each decision. The array `payoffs` holds the straddle payment owed at expiry. Each batch also carries its path count and dataset identity with its first path index. Registering disjoint ranges assigns the observations to training and validation before the tensors are uploaded.

Training consumes the uploaded training batch and advances the model version once per accepted update. Evaluation measures the current policy on the separate validation batch. Replacing or streaming training batches extends this loop to larger scenario sets. Multiple actors can submit work to one model using the same versioned update protocol. The supplied example implements retries after an actor receives a stale-version response.

For a current market observation, inference returns target stock and option holdings. With spot 100, normalized log spot zero and the full two-day horizon remaining, the input uses features $[0, 1]$ and zero previous holdings.

The returned `holdings` vector uses the stock and call ordering of the scenario marks. The application subtracts its current holdings to form orders. A checkpoint preserves model weights, Adam moments, configuration and dataset declarations. Restoring it reproduces holdings inference; continued training re-uploads the required scenario tensors. The client releases resources with `client.delete_native_model(model)`.

These calls map directly to HTTP: model creation uses `POST /v2/native/models`; the other operations use `POST /v2/native/models/{id}/{operation}`. Applications can use these routes from another language or register another compatible policy engine. The worker keeps the financial tensors and optimizer in its execution backend; the application receives holdings, risk summaries and versioned model state.

The request sequence below creates the learner, submits two scenario batches, trains it, evaluates held-out paths, retrieves holdings and restores a checkpoint. The premium and scenario batches are the inputs described above.

```
import math
from jormungandr.client import JormungandrClient
client = JormungandrClient("http://127.0.0.1:8080")
model = "hedge"
client.create_native_model({
  "model_id": model, "backend": "volt_burn",
  "config": {
    "target": "cpu", "precision": "f64",
    "premium": premium,
    "objective": {"kind": "squared_loss"},
    "trading_costs": [0.0005, 0.01],
    "liquidation_costs": [0.0005, 0.01],
    "cash_growth": [math.exp(0.03/252)] * 2,
    "income_growth": [math.exp(0.01/252), 1.] * 2})
client.native_command(model, "register_dataset", {
  "dataset_id": "gbm-study", "partitions": [
    {"split": "train", "first_path": 0, "paths": 64},
    {"split": "validation", "first_path": 64,
     "paths": 64}]})
for name, batch in [("train", training_batch),
                    ("validation", validation_batch)]:
    client.native_command(model, "upload", {
      "batch_id": name, "split": name, "batch": batch})
for _ in range(30):
    version = client.get_native_model(model)["version"]
    client.native_command(model, "update", {
      "batch_id": "train", "expected_version": version})
report = client.native_command(model, "evaluate", {
  "batch_id": "validation"})
targets = client.native_command(model, "infer", {
  "step": 0, "paths": 1, "features": [0., 1.],
  "previous_holdings": [0., 0.]})
state = client.native_command(model, "checkpoint")
state.pop("ok", None)
client.create_native_model({"model_id": "resumed",
  "backend": "volt_burn", "checkpoint": state})
```

6 Validation and measured results

We checked the trading calculation against a second implementation that follows each market path one date at a time using ordinary numbers and loops. It pays for holdings and fees, accrues interest and settles the liability. Both versions were written for this project; their separate code paths help catch sign errors, missing charges and incorrect payment timing. Parameter derivatives were also checked by perturbing weights and comparing the resulting change in loss.

The CPU double-precision checks agree to roughly 10^{-15} across the loss, parameter derivatives and first optimizer update. The corresponding CPU/Metal single-precision checks agree to roughly 10^{-6} for path profits. The validation records retain the individual errors and test inputs.

A companion comparison also executed the original Deep Hedging author's published financial engine and network components [6]. With matched inputs, both implementations reached validation CVaR of approximately 0.262 in a two-date Black–Scholes learning control after 1,000 updates.

The matched Heston application extends this check to 30 hedge dates, stock and a funded variance instrument. Three paired training seeds used the same initial weights, 131,072 training paths, 8,192 validation paths and 10,000 updates. On 32,768 fresh common paths, mean policy CVaR was 0.28700 for the author implementation and 0.28896 for Volt. The author-minus-Volt interval was $[-0.00828, 0.00463]$ at the adjusted 95% level. The model hedge produced 0.24552. Increasing both the training and validation pools substantially improved both learning engines relative to their earlier small-pool fits.

The learner also provides `volt_shared_lstm_reset`: two shared LSTM layers followed by two dense ReLU layers and linear holdings outputs. Its hidden state resets at each decision, following the reference implementation's one-observation sequence. Optional wealth inputs carry the financed account into the decision and remain connected through differentiation during training. Dropout has an explicit probability and seed. CPU and Metal/fusion checks cover forward holdings, financial loss, all parameter derivatives, the first Adam update and checkpoint continuation.

The end-to-end synthetic exercise used two trading days, a constant-volatility stock process with annual volatility 20%, interest 3% and dividend yield 1%. The liability was an at-the-money straddle and the hedge instruments were stock and an initially 84-day at-the-money call. Black–Scholes prices supplied the initial premium and hedge-call marks. Both networks had hidden width eight. Trading fees were 5 basis points for stock and 1% for the option, including terminal liquidation.

Two actors each supplied one batch of 64 training paths and applied 30 updates, for 60 accepted updates in total. Each actor reused its batch. On a separate 64-path validation set, mean squared hedging loss fell from 1.54 to 0.88, a 43% reduction. This run exercises data submission, shared-model updates and out-of-batch evaluation. Estimating economic performance calls for larger independent samples, repeated training runs and direct comparisons with model hedges.

The 63-day stock-and-option application uses joint return and implied-volatility transitions with the coefficients supplied by the reference example. Three shared-LSTM policies completed 200 updates each, with batches of 1,000 paths. Training used 32,000 paths and selection used 4,000 separate paths; assessment then used 8,000 fresh common paths. Mean squared hedging loss averaged 0.462, versus 3.077 for the source delta control, 2.121 for a delta–gamma hedge using pricing Greeks, and 24.750 without hedging. Mean fees were approximately 0.487 for the learned policies and 1.145 for delta–gamma hedging.

This initial comparison omits the source's learned small-trade rule and uses the stated finite training budget. All three selections occurred at update 200. The results demonstrate useful cost-aware learning for the supplied market specification. The original paper's longer training campaign and independently calibrated historical markets provide further comparisons. A 1,000-path, 63-date CPU preflight measured approximately 0.23–0.26 seconds per complete update; scenario generation and initial submission are additional costs.

CPU workload	Volt	PyTorch
64 paths, 2 dates, width 8	0.355	0.360
256 paths, 30 dates, width 17	5.957	5.440

Table 1: Median warm training-update time in milliseconds on an Apple M3 Max, macOS, float64, one CPU thread per runtime. Each update includes the trading calculation, backward differentiation and Adam optimization.

The timing comparison used five warm-up updates and 20 alternating measured updates. Volt used a tuned release build with ThinLTO and native CPU targeting. Its time includes worker communication; PyTorch runs in process. Initial upload and model creation are excluded. These measured workloads are small; the table provides an execution reference for the application calculation.

Application checks also cover dataset-partition enforcement, stale model versions, checkpoint continuation and a different registered learner engine. The completed release validation included 288 Volt checks, 216 Jormungandr checks and six native protocol/HTTP checks. These support the availability of the application interface and numerical engine. The controls support using the API to construct and compare hedging policies.

7 Using the workflow in a trading application

The same interfaces support a research comparison and a deployed decision process. For research, the application uploads common held-out market paths and evaluates the learned policy alongside delta and delta–gamma hedges, using the same premiums, instrument units, financing and execution costs. Loss, turnover and account paths then explain the economic effect of the policy. The mathematical market specification stays with the dataset so that the comparison can be reconstructed.

For ongoing decisions, the application prepares the observation using the training-time feature definitions, supplies current holdings and requests the next target. The step index selects the policy for the remaining horizon. Stock and option prices, contract maturity and any volatility features must use the same conventions as training. Portfolio quantities convert the targets into economic exposures; the execution application handles order sizes, liquidity and position controls.

Changing the liability or market model begins with a new scenario specification. Longer horizons, additional instruments and new market observations enter through the configured policy dimensions and batch arrays. Predictive factors can enter as features when their observation times and training distribution are defined. This gives practitioners a common way to study whether additional information or a different hedge instrument improves risk after trading costs.

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